

Ava Robillard

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EDUCATION

Master of Environmental Data Science, 4.00 GPA (June 2026)

University of California, Santa Barbara- Bren School of Environmental Science & Management

Related Courses: Databases and Data Management, Geospatial Analysis, Data Visualization and Communication, Machine Learning, Statistics for Environmental Data Science

Bachelor of Science in Plant Sciences, Minor in Data Science, 3.95 GPA (May 2025)

Cornell University, Ithaca, NY

Related Courses: Data Analysis in Ecology, GIS, Remote Sensing, Introduction to Computing: Design and Development, Linear Algebra, Probability Models, Urban Plants and Public Health

Leadership: Plant Science Peer Mentor, Cornell Botanical Gardens Horticultural Enterprises

Honors: Pi Alpha Xi National Honor Society for Horticulture

DATA SCIENCE PROJECT EXPERIENCE

Rincon's Bio Weaver Tool- From Disparate Data to Actionable Analysis (1/26-6/26)

Client: Rincon Consultants, Inc. | Master's Capstone Project

- Developed a data pipeline integrating multiple biological databases enabling automated retrieval and structuring of species occurrence data, reducing manual processing steps in the client's workflow
- Designed an interactive application that accepts user-defined project parameters and generates maps, visualizations, and analysis-ready tables for regulatory reporting and environmental assessments
- Collaborated with a 3-person capstone team using GitHub to implement version-controlled workflows and produce comprehensive technical documentation supporting independent client use

NYC Street Tree Spatial Analysis | Data Visualization & Communication Course Project (1/26-3/26)

- Built a data-driven infographic examining urban tree equity across NYC boroughs, joining NYC Open Data's 600,000-record Tree Census with American Community Survey and Census TIGER/Line spatial data
- Published reproducible code on GitHub designed for future integration with the 2025 NYC Tree Census
- Infographic selected for permanent display at the Bren School of Environmental Science & Management

PROFESSIONAL EXPERIENCE

Computational Biology REU Intern – National Science Foundation, Chattanooga, TN (5/24-7/24)

- Processed and analyzed Landsat satellite imagery in Google Earth Engine to develop an Urban Heat Island Index using normalized indices for vegetation greenness and land surface temperature
- Designed spatial data maps in ArcGIS Pro, revealing temporal trends to inform urban planning strategies
- Presented research at the ASPRS International Technical Symposium (2024): Spatio-temporal Analysis of Urban Heat Island Effect Using Remote Sensing, GIS, and Google Earth Engine

Herbarium Assistant – L. H. Bailey Herbarium, Cornell University, Ithaca, NY (10/23-5/25)

- Recorded taxonomic data from herbarium specimen labels into a MySQL curatorial database
- Imaged plant specimens, contributing to the objective of digitizing an 800,000-specimen collection

Research Assistant II – Evolution of Plant Form & Function Lab, Cornell University, Ithaca, NY (2/22-5/25)

- Conducted bioinformatics analyses, including plastome annotation and alignments, single nucleotide polymorphism (SNP) calling, and phylogenetic inference using Bash scripts
- Extracted and quantified DNA using gel electrophoresis and Qubit for perennial species of *Glycine*
- Organized seed stocks and germinated seedlings for DNA sequencing of genetic diversity
- Co-author on Botany 2023 presentation: B Landis, A Robillard, A Mahan, and JJ Doyle. Overcoming phylogenomic challenges of nested polyploidy events in *Glycine* using genome-wide data.

SKILLS

Languages: Python, R, SQL, Bash

Tools & Platforms: Git/GitHub, ArcGIS, Google Earth Engine, Quarto, Microsoft Office Suite, Affinity